AHOPPA: Agentic Humanoid Operation through Perception, Planning, and Action

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https://hebbian-robotics.github.io/AHOPPA/

Abstract: We study online humanoid operation with frontier multimodal models and report results obtained in the last few weeks following their release. AHOPPA connects model decisions to the fixed SONIC controller on a simulated Unitree G1. The model uses cameras, a map and a revisable plan to inspect four drawers, push open drawers shut by hand and return home through numerical head/wrist and locomotion commands. Physics continues during inference; a simulator evaluator checks completion. Across 64 scheduled episodes with four models at low and high reasoning effort, GPT-6 Astra completed 12 of 14 task attempts, taking 615 to 1492 s. The episodes show visual inspection, contact and return-home behavior without a task skill library. Falls, timeouts and incorrect completion claims expose limits in simulation.

Keywords: Humanoid control, multimodal agents, online evaluation

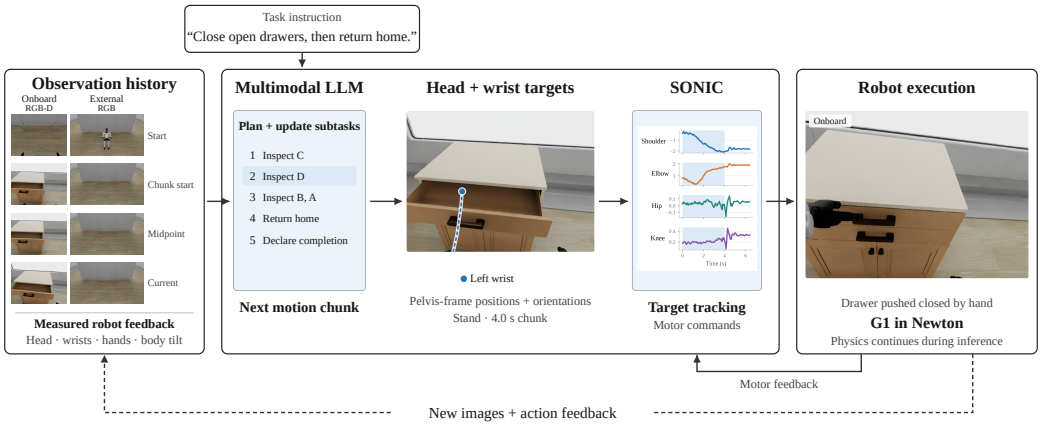


Figure 1: Visual feedback informs the model's plan and head/wrist targets with locomotion commands; SONIC tracks them while physics continues. Recorded inputs and commands are from a successful Astra (low) episode.

1 Online humanoid operation

We test whether frontier models can inspect cabinets, navigate, make contact and decide when to stop through visual feedback and numerical SONIC references [1].

Prior work uses grounded skills [2, 3, 4], code [5, 6], feedback [7, 8], spatial representations [9, 10, 11] or learned action tokens [12].

Model	Success	Physical	Closed	Seen	False	Control	Est. $	Episode (s)	Decision (s)
Astra (low)	6/7	6	9	5	0	1/1	9.30 / 10.57	817 / 926	9.2 / 9.4
Astra (high)	6/7	6	8	6	0	0/1	7.65 / 8.78	998 / 1145	21.4 / 21.8
Sol (low)	4/7	4	4	5	1	0/1	1.72 / 1.96	829 / 940	10.4 / 10.4
Sol (high)	3/7	5	9	6	0	0/1	1.74 / 1.59	1235 / 1162	21.5 / 21.8
Opus (low)	2/7	3	5	4	2	1/1	4.78 / 6.23	762 / 1028	13.2 / 13.5
Opus (high)	3/7	3	6	3	1	1/1	7.22 / 8.36	1392 / 1563	36.1 / 34.5
Gemini (low)	0/7	0	0	1	0	0/1	1.29 / –*	737 / –*	32.3 / –*
Gemini (high)	0/7	0	2	1	0	0/1	1.59 / –*	1332 / –*	136.7 / –*

Table 1: Seven scheduled task attempts per row; all-closed controls are separate. Other counts use completed episodes. *Physical*: closed, home, upright; *Seen*: all fronts viewed; *False*: declarations without physical completion. Pairs show all / successful attempts: mean cost and episode time, median per-attempt median decision latency. Costs use published rates [19, 20, 21]. *: no successful attempt observed. API-error starts stay in denominators.

2 Task, interface and evaluation

Scene. Newton [13] in Isaac Lab [14] runs at 200 Hz with SONIC at 50 Hz. Four Sektion cabinets [15] stand in NVIDIA's Simple Room [16, 17]: A and B behind 2.1 m lockers [18], C and D behind home. No drawer front is visible initially. Passive drawers close through contact, with 5 N dry friction and critically damped stops; the robot uses a flat ground collider. Eight conditions open zero, one or two drawers by 0.20 m under the same map and instruction.

Interface. Four head/wrist keyframes, walking/posture commands and binary hand presets drive SONIC (Fig. 1); facing stays within 15° of measured heading during turns. Decisions receive onboard RGB-D, external RGB, robot kinematics, depth queries, map, plan and feedback. Drawer joints, contacts and scores remain private; no task skills are provided.

Scoring. Closed means within 2 cm of shut and still for 1 s. Force sensors distinguish hand pushes from other contact. Success requires all drawers closed, a supported standing hold within 25 cm of home and the model's completion declaration. We separately record physical completion, disturbed drawers, contact attribution, falls, timeouts, infrastructure errors and all-closed controls.

Protocol. After pilots, we froze the setup, prompts, evaluator, source and 1800 s budget. GPT-6.1 Sol, GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon each ran one seeded episode per condition at low and high effort, four at a time with pinned host placement.

3 Results

Table 1 retains all 64 starts. All 43 closures were by hand; no initially closed drawer was disturbed. Astra succeeded in 6/7 tasks at each effort (615 to 1492 s), usually checking every drawer. Sol succeeded in 4/7 and 3/7, Opus 2/7 and 3/7, Gemini none. High effort lengthened decisions without consistent gains; eight episodes timed out. Fifteen of 51 completed task starts ended in falls, one before any model command; four declarations lacked physical completion. In this sweep, Astra high cost less despite more output tokens because cached reads covered 50% of its input, versus 29% at low.

Timing. Median peak physics lag was 192 ms. Only 17/58 completed episodes, including 7 successes, met the 100 ms limit; the results retain timing-limited episodes. Falls rarely followed lag spikes.

Limits. One simulated layout, drawer interaction and trial per model/effort/condition limit generalization. Friction, damping and ground support are modeling choices. Reliability, gains and transfer need repeated trials, older-model baselines and hardware experiments.

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